Types of Surveys - Detailed Notes

1. Plane Survey

Plane survey assumes the Earth's surface to be flat and neglects curvature. It is suitable for small areas, determining relative positions of points by measuring distances and angles for maps, boundaries, and construction projects.

2. Geodetic Survey

Geodetic survey considers the Earth's curvature to measure large areas accurately. It establishes widely spaced control points using triangulation and precise instruments, forming the basis for national mapping and higher-order control networks.

3. Engineering Survey

Engineering survey collects precise data for planning, designing, and constructing engineering projects such as highways, bridges, railways, dams, and buildings. It includes topographic surveys, earthwork measurement, grade setting, and as-built measurements.

4. Defence Survey

Defence survey provides critical information for military use. It prepares topographical and aerial maps showing enemy routes, airports, missile sites, and troop positions, supporting strategic and tactical planning for defence and attack operations.

5. Geological Survey

Geological survey studies surface and subsurface structures to locate and estimate mineral, oil, and rock reserves. It helps identify folds, faults, and rock types and guides foundation treatment for major engineering works.

6. Geographical Survey

Geographical survey gathers data to prepare maps showing natural and artificial features such as rivers, hills, forests, roads, and cities. It helps in land use planning, irrigation mapping, and understanding physical geography.

7. Mine Survey

Mine survey involves surface and underground measurements to map mine workings and determine the position of tunnels, shafts, and drifts. It ensures safe extraction of minerals and prepares geological and excavation plans.

8. Archaeological Survey

Archaeological survey locates, maps, and studies ancient sites, temples, forts, and relics buried underground. It aids in excavation, identification, and preservation of remains to understand the evolution and culture of past civilizations.

9. Land Survey

Land survey measures land boundaries, distances, and areas to prepare maps and property records. It includes setting monuments and locating points for topographical, cadastral, and city mapping.

10. Cadastral Survey

Cadastral survey determines and records land ownership boundaries and divisions. It is used for legal, taxation, and property documentation, covering both urban and rural plots.

11. City Survey

City survey collects detailed information about streets, buildings, and property lines in urban areas. It aids in city planning, property valuation, and maintenance of accurate municipal records.

12. Hydrographic Survey

Hydrographic survey measures and maps water bodies like rivers, lakes, and seas. It determines depths, tides, currents, and underwater topography for navigation, water supply, and marine engineering works.

13. Aerial Survey

Aerial survey uses aerial photographs or satellite imagery to map large or inaccessible areas. It's useful for topographic mapping, land development planning, and monitoring large-scale construction projects.

14. Chain Survey

Chain survey measures linear distances between points using chains or tapes. It's simple, economical, and suitable for small, open, and level areas without major obstacles.

15. Compass Survey

Compass survey determines directions and angles between lines using a magnetic compass. It's ideal for reconnaissance, route, and rough surveys where high precision isn't critical.

16. Leveling Survey

Leveling survey measures elevation differences between points to determine heights and gradients. It's used for contour mapping, road design, and establishing levels for construction and water flow.

17. Triangulation Survey

Triangulation survey forms a network of triangles to calculate distances and positions. Measuring one baseline and the included angles provides precise control points for large-scale and geodetic surveys.

18. Tacheometric Survey

Tacheometric survey determines horizontal and vertical distances indirectly using a theodolite with a stadia diaphragm. It's faster than leveling and ideal for contouring, reservoirs, and hilly or inaccessible terrain.